

Long Nguyen Khac

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Education

Vietnam National University, Hanoi – International School

Hanoi, Vietnam

B.Eng. in Automation and Informatics

Sep. 2021 – Dec. 2026

- **GPA:** 3.71/4.00; ranked in the top 4% of the cohort.
- Completed a one-year gap period for international research and exchange programs in Indonesia, Taiwan and Singapore.
- **Research focus:** robotics, wireless power transfer, and AI.

Research Experience

Water & Energy Research Lab, School of EEE, Nanyang Technological University

Singapore

Research Fellow

Feb. 2026 – Mar. 2026

- Conducted research on relay-coil-enhanced spatial wireless power transfer systems for autonomous underwater vehicle charging.
- Analyzed coil configuration, power transfer efficiency, and system-level integration for docking-free underwater charging.
- Explored mechanical and communication requirements for integrating wireless charging into long-endurance AUV platforms.

AISC Lab, AIM-HI Institute, National Chung Cheng University

Chiayi, Taiwan / Remote

Research Intern / Remote Research Student

May 2025 – Present

- Developed data-driven modeling approaches for cutting force prediction in CNC milling using large-scale experimental datasets.
- Applied nonlinear system analysis, sparse identification, and machine learning methods to model machining dynamics.
- Built data processing and model evaluation pipelines for machining force prediction under different experimental conditions.

Intelligent Control and Artificial Intelligence Laboratory, VNUIS

Hanoi, Vietnam

Student Researcher

Sep. 2022 – Present

- Currently conducting research on adaptive control for autonomous underwater vehicles, focusing on trajectory tracking under model uncertainties and external disturbances.
- Designed the mechanical model and physical prototype of a 3-DOF delta parallel robot, and implemented adaptive fuzzy backstepping control for robust trajectory tracking under uncertainties.
- Built a dataset of 2,632 construction-site images and trained YOLOv5s-based models for personal protective equipment detection.

Publications

1. **Long Nguyen Khac** et al. *Adaptive Fuzzy Logic in Backstepping Control for 3-DOF Parallel Robot*. International Conference on Advances in Information and Communication Technology, Springer, 2024. DOI: 10.1007/978-3-031-80943-9_57.
 - **Main highlights:** adaptive fuzzy backstepping control for robust trajectory tracking of a 3-DOF delta parallel robot.
 - **Presentation:** Mechatronics and Automation section, The 3rd International Conference ICTA 2024.
2. **Long Nguyen Khac** et al. *YOLOv5s Model with Applied Activation Functions for Personal Protective Equipment Detection in Construction Sites*. Proceedings of the National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability, 2023.
 - **Main highlights:** YOLOv5s-based PPE detection using six activation functions and a dataset of 2,632 construction-site images.
 - **Presentation:** National Conference on Smart Technology Applications in Industry 4.0, Smart City, and Sustainability, 2023.

International Academic Experience

School of Electrical and Electronic Engineering, Nanyang Technological University

Singapore

Global Connect Fellow

Feb. 2026 – Mar. 2026

- Joined an international research environment in electrical and electronic engineering, with exposure to interdisciplinary work across energy systems, robotics, and autonomous systems.

College of Design and Engineering, National University of Singapore

Singapore

Exchange Student

Aug. 2025 – Dec. 2025

- Completed a one-semester academic exchange, gaining broader exposure to engineering education, international research culture, and interdisciplinary learning.

College of Engineering, National Chung Cheng University

Taiwan

Research Intern

May 2025 – Jul. 2025

- Worked in an international research setting at AIM-HI Institute, developing experience in smart manufacturing, data-driven modeling, and collaborative research practice.

ASEAN in Today's World, Kyushu University

Indonesia

VNU representative

Feb. 2025

- Participated in an ASEAN-focused academic program, strengthening regional understanding, cross-cultural communication, and awareness of science, technology, and society in Southeast Asia.

Temasek Foundation Specialist's Community Action & Leadership Exchange

Singapore

VNU Representative

Feb. 2024

- Engaged in a regional leadership and community action program, developing teamwork, communication, and social responsibility in an international environment.

Overseas Immersion Programme

Vietnam

VNUIS Representative

Feb. 2023

- Engaged in an international immersion program with students from Ngee Ann Polytechnic, Singapore, developing cross-cultural communication, teamwork, and exposure to applied learning in a regional context.

Selected Honors & Awards

Fully Funded Scholarship, NUS Enterprise Summer Programme in Entrepreneurship, NUS ASEAN SPARK Fellowship	Jul. 2026
Most Popular Poster Award, NTU Singapore Global Connect Fellowship Final Research Poster Presentation	Mar. 2026
Exchange Scholarship, Kyushu University Spring-Semester Exchange Scholarship, Department of Mechanical Engineering	Jan. 2026
Research Fellowship, Selected as one of 39 young scholars worldwide for NTU Singapore Global Connect Fellowship	Feb. 2026
Fully Funded Scholarship, DiscoverNUS Scholarship, National University of Singapore	Aug. 2025
Research Internship Scholarship, Taiwan Experience Education Program, Ministry of Education	May. 2025
International Program, ASEAN in Today's World, Kyushu University	Feb. 2025
Scholarship, TOSHIBA Scholarship, TOSHIBA International Foundation	Sep. 2024
Fully Funded Exchange Program, Temasek Foundation Specialist's Community Action & Leadership Exchange	Feb. 2024
Fighting Spirit Prize, South-East Asia Circuit and System Society Hackathon	Dec. 2022
Academic Scholarship, Six-semester recipient for excellent academic performance	2021-2025

Selected Teaching Experience

Faculty of Engineering and Technology, VNUIS

Hanoi, Vietnam

Teaching Assistant

Sep. 2023 – Dec. 2024

- **Modeling and Simulation of Control Systems**
 - Assisted 30 students across 7 laboratory sessions on MATLAB/Simulink for robotics, mechatronics, and process control applications.
 - Supported the preparation of laboratory manuals, tutorial exercises, and simulation-based learning materials.
- **Microprocessors and Microcontrollers**
 - Mentored 30 students through 15 laboratory sessions on Embedded C, Arduino, STM32, and embedded system development.

Selected Working Experience

ETEK Automation Solutions JSC

Hanoi, Vietnam

Engineering Intern

Mar. 2025 – May 2025

- Project: AGV-based Virtual Simulation for Automated Warehouse Logistics – VINFAST Hai Phong.
- Sketched AGV/AMR traffic intersections using AutoCAD and simulated warehouse logistics operations using Visual Components Premium 4.6.
- Monitored AGV/AMR behavior, identified simulation errors, and supported troubleshooting of automated logistics workflows.

Vietnam E-Commerce and Digital Economy Agency, Ministry of Industry and Trade

Hanoi, Vietnam

Technical Collaborator

Jul. 2023 – Sep. 2023

- Project: Consultancy Support and Sales Policy Development for E-Commerce Website.
- Coordinated with SMEs across Dak Nong and An Giang provinces to support e-commerce platform development.
- Assisted in website design, content management, and preparation of technical manual documents.

Skills

Programming	MATLAB, Python, C/C++
Control	Mathematical modeling, MATLAB/Simulink, PID control, adaptive control, trajectory tracking
Robotics	AUV systems, Delta parallel robots, AGV/AMR simulation, mechanical design
AI/Data	Machine learning, deep learning, data-driven modeling, computer vision
Embedded Systems	STM32, Arduino, PLC programming, embedded system integration
Tools	SolidWorks, AutoCAD, Visual Components, PyTorch, TIA Portal
Languages	Vietnamese: Native; English: Fluent, Chinese: Beginner, French: Beginner

References

- Asst. Prof. Yun Yang** — School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore. **Email:** yun.yang@ntu.edu.sg
- Prof. Her-Terng Yau** — Advanced Institute of Manufacturing with High-tech Innovations, National Chung Cheng University, Taiwan. **Email:** yauhertern@gmail.com
- Dr. Le Xuan Hai** — International School, Vietnam National University, Hanoi **Email:** hailx@vnu.edu.vn